



Republic of Rwanda  
Ministry of Education



**RTB** | RWANDA  
TVET BOARD

## QUANTIFYING CONSTRUCTION WORKS

**BDCQC401**

**Quantify Construction works**

### Competence

RQF Level: 4

Learning Hours



Credits: 5

Sector: Construction and Building Services

Trade: Building Construction

Module Type: Specific

Curriculum: CBSBDC4001- TVET Certificate 4 in Building Construction

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|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  |                      |                      |     |                   |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|----------------------|----------------------|-----|-------------------|
| <b>Purpose statement</b>               | This module describes the knowledge, skills and attitude of quantifying various construction works and reporting the work done. It is intended to be pursued by the learners at TVET Certificate IV in building construction. At the end of this module, the learner will be able to: Interpret construction drawings, determine required resources in the construction sites, perform site data collection and reporting. |     |  |                      |                      |     |                   |
| <b>Learning assumed to be in place</b> | Building construction drawing; Apply fundamental mathematical analysis, Basics of AutoCAD, Cement based blocks and pavers works, Concrete works, Reinforcements steel bars works, Terrazzo works, Tile works.                                                                                                                                                                                                              |     |  |                      |                      |     |                   |
| <b>Delivery modality</b>               | <b>Training delivery</b>                                                                                                                                                                                                                                                                                                                                                                                                   |     |  | <b>100%</b>          | <b>Assessment</b>    |     | <b>Total 100%</b> |
|                                        | Theoretical content                                                                                                                                                                                                                                                                                                                                                                                                        |     |  | 30%                  | Formative assessment | 30% | 50%               |
|                                        | Practical work:                                                                                                                                                                                                                                                                                                                                                                                                            |     |  | 70%                  |                      | 70% |                   |
|                                        | • Group project and presentation                                                                                                                                                                                                                                                                                                                                                                                           | 20% |  |                      |                      |     |                   |
|                                        | • Individual project /Work                                                                                                                                                                                                                                                                                                                                                                                                 | 50% |  |                      |                      |     |                   |
|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                            |     |  | Summative assessment |                      |     | 50%               |

### Elements of competency and performance criteria

| Elements of competency                     | Performance criteria                                                                            |
|--------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>1. Interpret construction drawings.</b> | 1.1. Types of lines are correctly interpreted according to their use.                           |
|                                            | 1.2. Dimensions of drawing are accurately interpreted with respect to their use.                |
|                                            | 1.3. Drawing symbols and signs are correctly interpreted with respect to their use.             |
|                                            | 1.4. Drawing scale is accurately interpreted in accordance to the occupation standard.          |
|                                            | 1.5. Construction drawing plans are systematically interpreted with respect to the standard.    |
|                                            | 1.6. Construction drawing sections are systematically interpreted with respect to the standard. |
|                                            | 1.7. Construction drawing views are systematically interpreted with respect to the standard.    |

|                                                              |                                                                                                   |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|                                                              |                                                                                                   |
| <b>2. Determine required resources in construction site.</b> | 2.1. Units of measurements are correctly identified in accordance to the international standards. |
|                                                              | 2.2. Building materials are precisely quantified in accordance to the work to be done.            |
|                                                              | 2.3. Human resource is correctly determined in accordance to the work to be done.                 |
|                                                              | 2.4. Building components breakdown is properly done according to the type of work.                |
| <b>3. Perform site data collection and reporting.</b>        | 3.1. Site data to be recorded are correctly identified according to the building structure.       |
|                                                              | 3.2. Site data forms are properly filled in accordance to the type of work.                       |
|                                                              | 3.3. The work is correctly measured based on its location                                         |
|                                                              | 3.4. Site data are correctly recorded according to the format                                     |
|                                                              | 3.5. Collected site data are correctly reported according to the reporting format                 |

### Intended Knowledge, Skills and Attitude

| Knowledge                                                                                                                                                                                                                | Skills                                                                                                                                                                                                                                                                                                                                   | Attitude                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>✓ Quantify building materials from measured works and construction drawings</li> <li>✓ Demonstrate Strong grammar and spelling while recording site data and reporting</li> </ul> | <ul style="list-style-type: none"> <li>✓ Apply communication skills at workplace</li> <li>✓ Apply Safety Precautions Measures</li> <li>✓ Interpret Drawings</li> <li>✓ Collect Site Data and Report the work done</li> <li>✓ Select and use construction measuring tools and equipment</li> <li>✓ Recording resources used on</li> </ul> | <ul style="list-style-type: none"> <li>✓ Demonstrate concentration and attention to details while receiving instruction, taking measurements, drawing interpretation, work quantification and reporting.</li> <li>✓ Demonstrate persistent, team working, flexibility, punctuality during measurement work quantification, and reporting</li> <li>✓ Demonstrate the ability to</li> </ul> |

|  |                                                                                                                                                                                                                                                                              |                                                                                                                                                                                             |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | site on a regular basis<br>✓ Apply resource Management skills<br>✓ Apply innovative and problem-solving skills<br>✓ Apply reading and writing skills during measurements taking and drawing interpretation<br>✓ Apply Competent keyboard skills<br>✓ Apply leadership skills | Work independently with minimal supervision<br>✓ Respect time assigned to every task<br>✓ Being stress tolerant and self-motivated<br>✓ Demonstrate risk awareness by putting on proper PPE |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Course content

|                                                            |                                                                                                                                                                                                                                          |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Learning outcomes</b>                                   | <b>At the end of the module the learner will be able to:</b><br><br><b>1. Interpret construction drawings</b><br><b>2. Determine required resources in the construction site</b><br><b>3. Perform site data collection and reporting</b> |
| <b>Learning outcome 1: Interpret construction drawings</b> | <b>Learning hours: 10</b>                                                                                                                                                                                                                |

### Indicative content

- **Interpretation of drawing lines**
  - Types of drawing line
  - Application of drawing line
- **Interpretation of drawing dimensions**
  - Types of dimensions
    - ✓ Size dimensions
    - ✓ Location dimensions
  - Method of dimensioning
    - ✓ Aligned method
    - ✓ Unidirectional method
  - Elements of dimensioning
    - ✓ Dimension line
    - ✓ Leader

- ✓ Extension line
  - Units of dimensioning
- **Interpretation of drawing symbols and signs**
  - Types of symbols and signs used in construction drawing
  - Functions of symbols and signs for construction drawing
- **Interpretation of drawing scales**
  - Sizes of scales
    - ✓ Full size scales
    - ✓ Reducing scales
    - ✓ Enlarging scales
  - Scale representation
    - ✓ Fractional Representation
    - ✓ Ratio representation
    - ✓ Graphical representation
- **Interpretation of construction drawing plans**
  - Foundation plan
  - Floor plan
  - Roof plan
  - Elevation
  - Site plan
  - Location plan
- **Interpretation of construction drawing sections**
- **Interpretation of construction drawing views**
  - Front view
  - Back view
  - Top view
  - Left side view
  - Right side view

#### Resources required for the learning outcome

|                              |                                                                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Equipment                    | PPEs, calculator, laptop                                                                                                                         |
| Materials                    | Construction drawing handout, Pencil lead, Lead, A4 Papers, highlighters                                                                         |
| Tools                        | -Compass, ruler                                                                                                                                  |
| Facilitation techniques      | <ul style="list-style-type: none"> <li>• Group discussion</li> <li>• Trainer guided</li> <li>• Demonstration</li> <li>• Brainstorming</li> </ul> |
| Formative assessment methods | <ul style="list-style-type: none"> <li>• Written assessment</li> <li>• Performance assessment</li> </ul>                                         |

- Matching questions
- Multiple choice
- Sentence completion
- Short questions
- Open ended questions

**Learning outcome 2: Determine required resources in the construction site**

**Learning hours:30**

**Indicative content**

- **Identification of measurements units**
  - Type of units of measurement
    - ✓ Single unit
    - ✓ Linear measurement
    - ✓ Real surface measurement
    - ✓ Cubic content
  - Principles of deciding units of measurement
  - Rules for measurement
- **Quantification of building materials**
  - Identification of building materials
    - ✓ Types of building materials
    - ✓ Specification of building materials
  - Methods of calculating the quantity of materials
    - ✓ Long wall-short wall method
    - ✓ Centre line method.
    - ✓ Crossing method.
    - ✓ Bay method
    - ✓ Service unit method
  - Quantify the number of building materials
    - ✓ Cement
    - ✓ Tiles
    - ✓ Bricks
    - ✓ Sand
    - ✓ Gravel
    - ✓ Sheet metals
- **Determination of human resource**
  - Factors for human resources selection
  - Labour classification
- **Breakdown of building components**
  - Foundation

- Footing
- Wall
- Column
- Beam/lintel
- Slab
- Pavement/floor
- Roof

#### Resources required for the indicative content

|                              |                                                                                                                                                                                                                                                             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Equipment                    | – PPEs, measuring instrument, computer/laptops, calculator, dumpy level<br>DGPS/Total station                                                                                                                                                               |
| Materials                    | -Pen, pencil, highlighters, papers, notebook, marking chalk, marker pen                                                                                                                                                                                     |
| Tools                        | Measuring tapes, vernier caliper, compass, decameter                                                                                                                                                                                                        |
| Facilitation techniques      | <ul style="list-style-type: none"> <li>• Group discussion</li> <li>• Trainer guided</li> <li>• Brainstorming</li> </ul>                                                                                                                                     |
| Formative assessment methods | <ul style="list-style-type: none"> <li>• Written assessment</li> <li>• Performance assessment</li> <li>• Matching questions</li> <li>• Multiple choice</li> <li>• Sentence completion</li> <li>• Short questions</li> <li>• Open ended questions</li> </ul> |

#### Learning outcome 3: Perform site data collection and reporting

Learning hours: 10

#### Indicative content

- **Identification of data to be recorded on the site**
  - Work done on the site
  - Labour
  - Materials used
- **Fill site forms**
  - Types of site documents/books
  - Methods of document arrangement
  - Safeguard site documents/book
  - Types of forms
- **Measure the work done**
- **Record the work done**
  - Benefits of recording data

- **Report the work done**

- Content of site report format
- Parts of site report format
- Important of site report
- Process of reporting data to the supervisor

### Resources required for the indicative content

|                              |                                                                                                                                                                                                                                                             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Equipment                    | PPEs, Measuring instruments, Computer/laptops, Calculator, Dumpy level DGPS/Total station.                                                                                                                                                                  |
| Materials                    | -Pen, Pencil, Highlighters, Papers, Notebook, Marking chalk.                                                                                                                                                                                                |
| Tools                        | Measuring tapes, Vernier caliper, Compass, Decameter.                                                                                                                                                                                                       |
| Facilitation techniques      | <ul style="list-style-type: none"> <li>• Group discussion</li> <li>• Trainer guided</li> <li>• Brainstorming</li> </ul>                                                                                                                                     |
| Formative assessment methods | <ul style="list-style-type: none"> <li>• Written assessment</li> <li>• Performance assessment</li> <li>• Matching questions</li> <li>• Multiple choice</li> <li>• Sentence completion</li> <li>• Short questions</li> <li>• Open ended questions</li> </ul> |

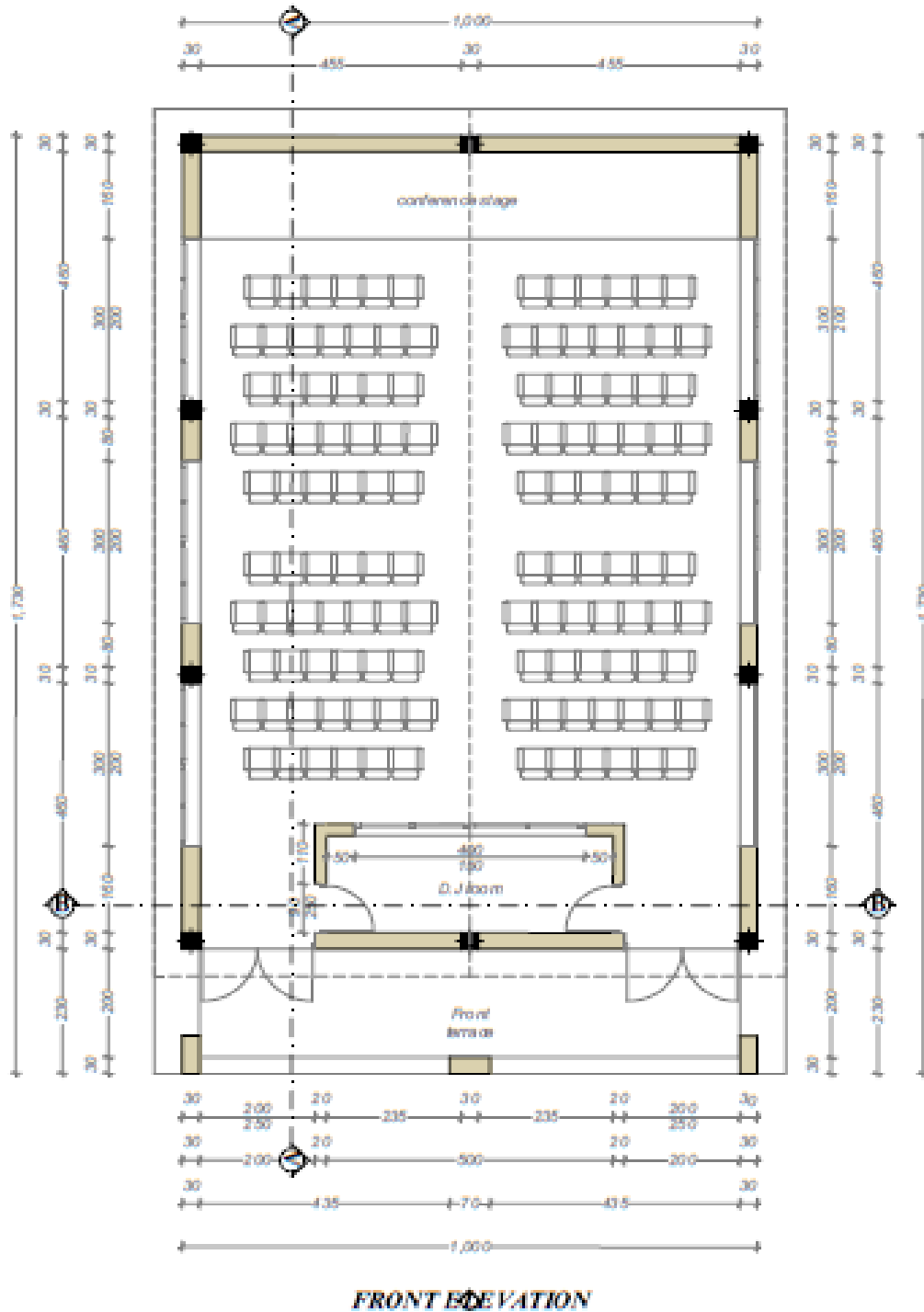
### Integrated/Summative assessment (For specific module)

#### Integrated situation

Musanze district has contracted JB company Ltd to construct its new conference hall. As a technician in Level 4 Building construction, you are requested to use the provided drawings and respect given dimensions to quantify the following tasks:

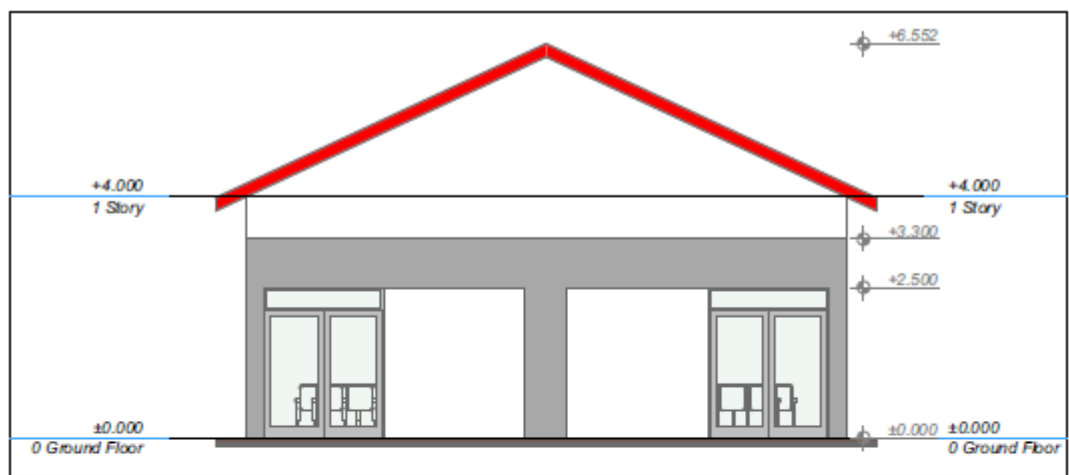
1. The volume of stone masonry in foundation
2. The volume of brick masonry in wall elevation
3. The flooring area of the building
4. The number of bricks required
5. The number of cement bags required
6. The volume of sand required
7. The volume of gravel required
8. The volume of concrete lintels
9. The number of iron sheet metal required
10. The number of doors
11. The number of windows
12. Report the work done

**LEFT ELEVATION**

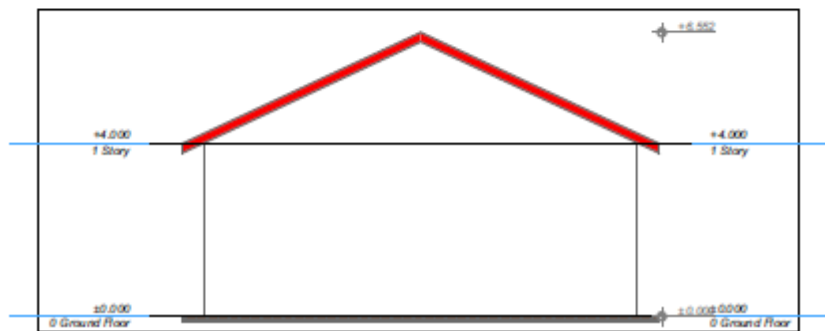


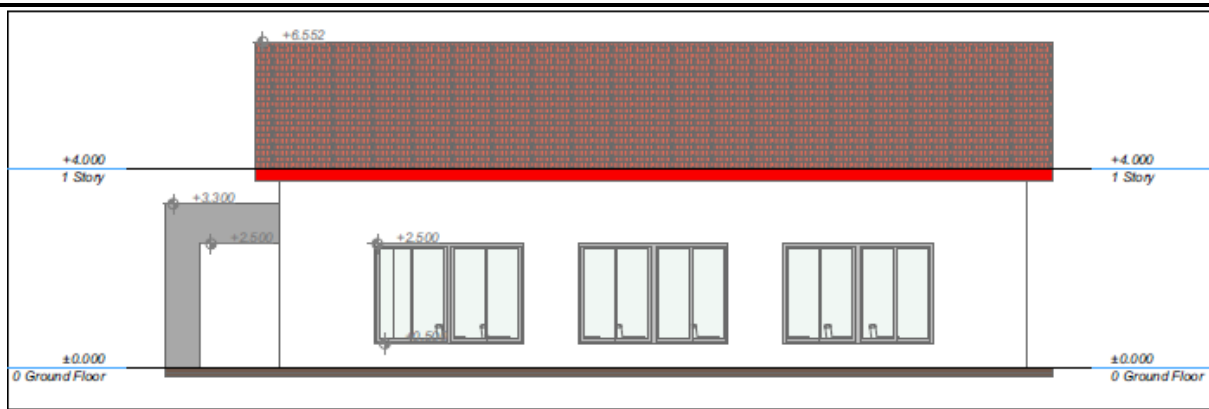
**RIGHT ELEVATION**

20m\*20m and .9 m and brick spectively, the

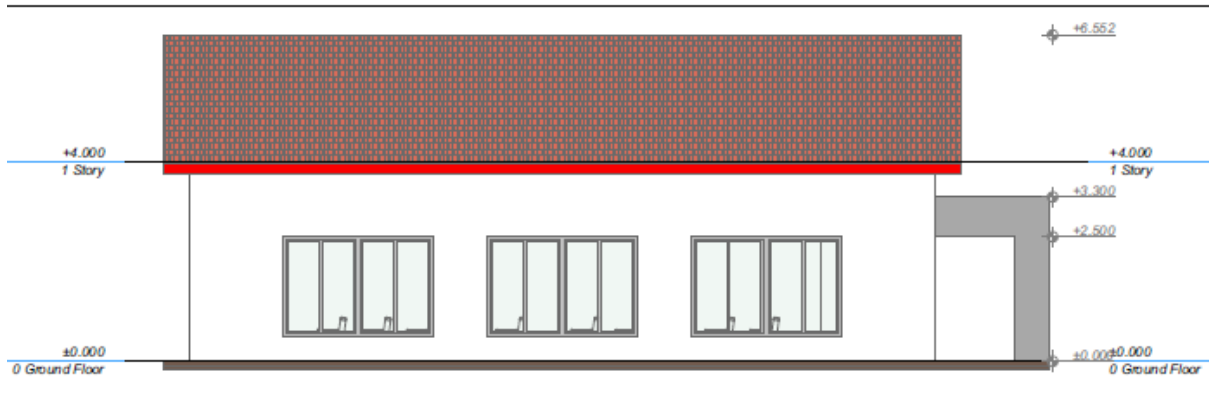


FRONT ELEVATION

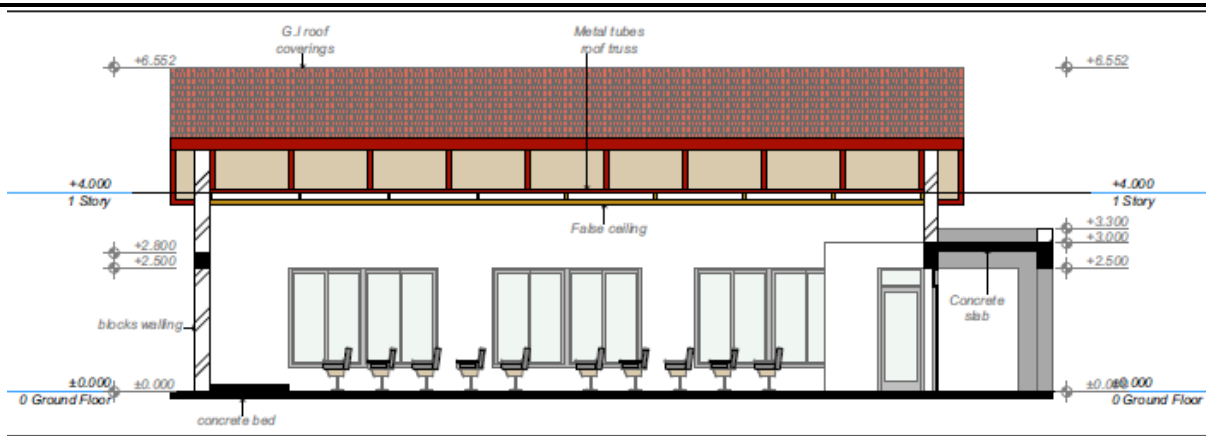




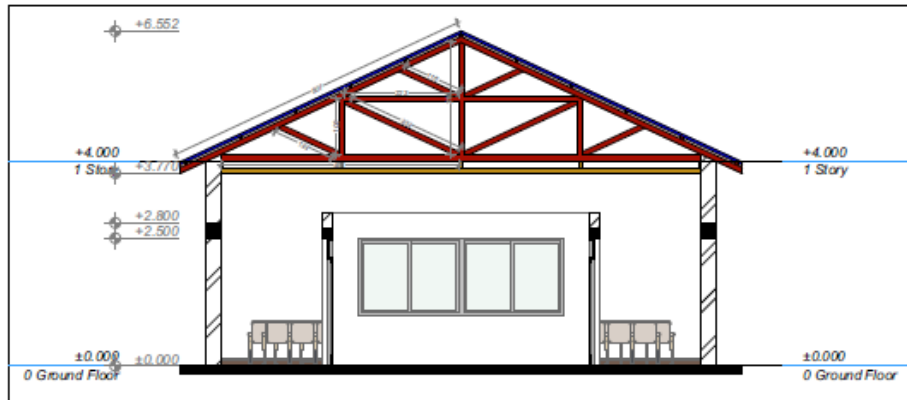
RIGHT ELEVATION



LEFT ELEVATION



LONGITUDINAL SECTION A-A



CROSS SECTION B-B

| Resources                                                                   |                                                                                                  |                                          |             |    |                  |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------|-------------|----|------------------|
| Tools                                                                       |                                                                                                  | Rulers, Compass                          |             |    |                  |
| Equipment                                                                   |                                                                                                  | Calculator                               |             |    |                  |
| Materials/ Consumables                                                      |                                                                                                  | Papers, drawing sheet, Pen, Pencil       |             |    |                  |
| Assessable outcomes                                                         | Assessment criteria (Based on performance criteria)                                              | Indicator                                | Observation |    | Marks allocation |
|                                                                             |                                                                                                  |                                          | Yes         | No |                  |
| Learning outcome 1: Interpret construction drawings (20%)                   | 1.1. Dimensions of drawing are accurately interpreted with respect to their use                  | Length is interpreted                    |             |    | 1                |
|                                                                             |                                                                                                  | Width is interpreted                     |             |    | 1                |
|                                                                             |                                                                                                  | Height is interpreted                    |             |    | 1                |
|                                                                             |                                                                                                  | Thickness is interpreted                 |             |    | 1                |
|                                                                             | 1.2 Drawing symbols and signs are correctly interpreted with respect to their use                | Concrete symbol is interpreted           |             |    | 2                |
|                                                                             |                                                                                                  | Brick symbol is interpreted              |             |    | 2                |
|                                                                             |                                                                                                  | Stone masonry symbol is interpreted      |             |    | 2                |
|                                                                             |                                                                                                  | Window symbol is interpreted             |             |    | 1                |
|                                                                             |                                                                                                  | Door symbol is interpreted               |             |    | 1                |
|                                                                             | 1.3. Construction drawing plans are systematically interpreted with respect to the standard      | Floor plan is interpreted                |             |    | 2                |
|                                                                             |                                                                                                  | Elevation is interpreted                 |             |    | 2                |
|                                                                             |                                                                                                  | Section is interpreted                   |             |    | 2                |
|                                                                             |                                                                                                  | Roof plan is interpreted                 |             |    | 2                |
| Learning outcome 2: Determine required resources in construction site (60%) | 2.1. Units of measurements are correctly identified in accordance to the international standards | Single unit of measurement is identified |             |    | 2                |
|                                                                             |                                                                                                  | Single unit of measurement is identified |             |    | 2                |
|                                                                             |                                                                                                  | Linear measurement is identified         |             |    | 2                |
|                                                                             |                                                                                                  | Cubic content measurement is identified  |             |    | 2                |
|                                                                             |                                                                                                  | Real surface measurement is identified   |             |    | 2                |
|                                                                             | 2.2. Building materials are precisely quantified in accordance to the work to be done            | The number of bricks is quantified       |             |    | 7                |
|                                                                             |                                                                                                  | The number of cement bags is quantified  |             |    | 7                |
|                                                                             |                                                                                                  | The volume of sand is quantified         |             |    | 7                |

|                                                                             |                                                                                   |                                              |  |  |   |
|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------|--|--|---|
|                                                                             |                                                                                   | The volume of gravel is quantified           |  |  | 7 |
|                                                                             |                                                                                   | The number of iron sheet metal is quantified |  |  | 7 |
|                                                                             | 2.4. Building components breakdown is properly done according to the type of work | Breakdown of foundation is done              |  |  | 3 |
|                                                                             |                                                                                   | Breakdown of footing is done                 |  |  | 3 |
|                                                                             |                                                                                   | Breakdown of wall is done                    |  |  | 3 |
|                                                                             |                                                                                   | Breakdown of pavement/floor is done          |  |  | 3 |
|                                                                             |                                                                                   | Breakdown of roof is done                    |  |  | 3 |
| <b>Learning outcome 3: Perform site data collection and reporting (20%)</b> | 3.3. The work is correctly measured based on its location                         | Data is collected from drawing               |  |  | 8 |
|                                                                             | 3.4.Site data are correctly recorded according to the format                      | Data is recorded to the format               |  |  | 5 |
|                                                                             | 3.5. Collected site data are correctly reported according to the reporting format | Work done is reported to the trainer         |  |  | 7 |
| <b>Total marks</b>                                                          |                                                                                   | <b>100</b>                                   |  |  |   |
| <b>Percentage Weightage</b>                                                 |                                                                                   | <b>100%</b>                                  |  |  |   |
| <b>Minimum Passing line % (Aggregate): 70%</b>                              |                                                                                   |                                              |  |  |   |

## References:

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2. Ivor H. Seeley. 2019 “Advanced Building Measurements 3rd Edition” . Nottingham
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